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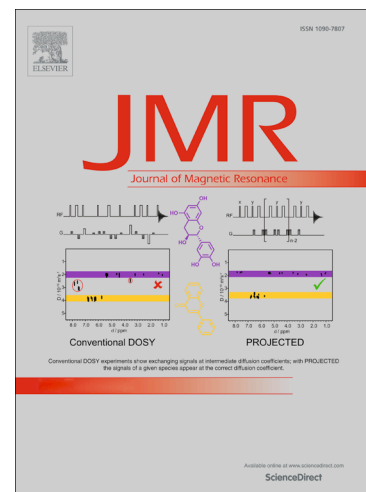
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A Resistive Q-switch for Low-field NMR Systems

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Abstract

A NMR Q-switch was designed and constructed specifically for use with low-field NMR apparatus. This featured a comparatively simple resistive damping design. It served to reduce the r.f. probe ring-down time, and hence reduced the signal acquisition delay from 25 ms to 9 ms, on an Earth's magnetic field NMR system. The advantage of this earlier acquisition was demonstrated for both an aqueous suspension of iron oxide particles and using an NMR flow meter.

Keywords: Q-switch; Ring-down; Damping; Low-field NMR; Earth's field NMR; Flow meter.

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1. Introduction

Ring-down in a nuclear magnetic resonance (NMR) acquisition system is unwanted voltage oscillation in the excitation/receiver probe due to residual stored energy immediately after the excitation radio frequency (r.f.) pulse is terminated. It is related to the quality factor (Q) of the probe and can be reduced by using various damping techniques [1-6]. Q is an effective efficiency measure given by the ratio of probe impedance at the operating frequency (ω_0) to its internal resistance, mainly from the inductor: $\omega_0 L/R$. Generally, systems with a higher Q achieve a better signal-to-noise ratio (SNR, which is proportional to $\sqrt{Q}$), but also present the drawback of a more pronounced ring-down. The ring-down time period is conventionally defined as a factor of the time constant, τ , where τ is determined as [3,4]:

$$\tau = 2Q/\omega_0 \quad (1)$$

For high resonance frequency (ω_0) NMR spectrometers, as widely used with superconducting magnets, this ring-down period is generally insignificant given both the inverse dependency on resonance frequency in Equation 1 and the larger signal intensity. However when dealing with comparatively low frequency spectrometers, the smaller frequency coupled with generally lower signal intensities mean that ring-down effects are more prominent.

A Q-switch helps to shorten the ring-down time and hence allows earlier signal acquisition by switching on a load to lower the Q of the system and hence speed up the energy dissipation process. The work reported here details the design and implementation of a Q-switch for an Earth's magnetic field NMR spectrometer [7]. This is a portable spectrometer which is capable of 2D and 3D imaging; whilst it is marketed primarily as a NMR teaching system, it is also capable of potential industrial applications using T_1 and T_2 measurements [8-10]. The Q-switch implemented here is a resistive damping design which has no frequency tuning components. While the work presented is demonstrated at ~ 2 kHz, corresponding to the Earth's magnetic field, the design is equally applicable to other systems operating below approximately 2 MHz.

The performance of the resultant Q-switch in improving the quality of the acquired NMR data is demonstrated using two case studies. The first considers NMR free induction decay (FID) measurements for suspensions of iron oxide particles of various concentrations. In a very general sense, such suspensions (albeit typically for superparamagnetic nanoparticles ($\sim < 30$ nm)) are widely considered in terms of point-of-care benchtop NMR devices for detecting labelled biomarkers for a suite of medical diagnostic purposes [11]. The second case study considers the use of the earth's magnetic field apparatus

as a flow meter [⁸⁻¹⁰]. This requires analysis of the FID; the previous inability to detect this at earlier times following the excitation r.f. pulse limited the range of velocities that could be detected due largely to flush-out effects.

2. Equipment Development

The design of the Q-switch focused on ease of construction and use – it was designed to be external to the Earth's magnetic field spectrometer used (the Terranova [⁷]), in line with the r.f. probe and thus not require modification to the existing hardware. A resistive damping method was preferred as it does not use tuning and matching elements, therefore it is independent of the probe operating frequency, further simplifying the hardware construction. An NMR probe is essentially a r.f. power transmitter and signal receiving antenna, and can be regarded as a LC resonant circuit. In the case of our design, L is the inductance of the B_1 coil/probe and C is the spectrometer capacitance. L and C thus form a parallel resonant circuit. Q damping switches on a resistive load immediately after a r.f. pulse, in the process lowering the Q of the system and hence speeding up energy dissipation from the LC circuit. A simplified schematic of the design is shown in Figure 1.

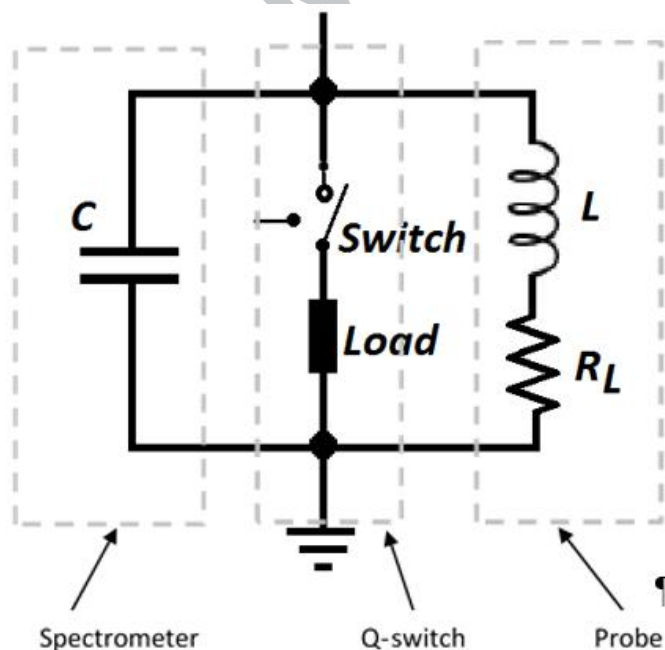


Figure 1: Q-switch for the Terranova system. C is ‘inside’ the Terranova spectrometer and adjustable using Magritek Prospa software and L is the B_1 coil in the probe with internal resistance R_L . The Load is the damping resistor that determines the new Q and hence the ring-down time of the system.

The resistive load needs to be rapidly switched on after the excitation r.f. pulse ends. Mechanical switches are not suitable as the reaction time and the contact de-bouncing takes tens of milliseconds and the ring-down time of the Terranova probe is approximately 25 ms. MOSFET [12] switching devices were used as they present switching times less than 1 ms. When using a MOSFET switch for r.f. signal in this context, the body diode of the MOSFET needs to be ‘off’ to prevent shorting of one of the cycles to ground if the r.f. signal is greater than the diode’s forward bias voltage (~ 0.7 V). To accommodate this requirement, two MOSFET switches were employed in a “back to back” configuration with the source terminals connected together. The resultant two body diodes prevent each other from conducting and the MOSFET switches can then be activated simultaneously. The common source also simplifies the MOSFET driving circuit as it provides a common point of reference for the required Gate-Source voltage. The MOSFETs were selected to have a low parasitic capacitance to minimize the effect on the LC tune circuit. A pair of SUD06 series N-channel MOSFETs from Vishay Siliconix were used [13]; the switching and recovery time of the devices is around 52 ns. Each MOSFET has a parasitic capacitance value of approximately 0.3 nF. The Terranova system used operates around 2.2 kHz, the tuning capacitance is approximately 10 nF. In using the Q-switch, this tuning capacitance was adjusted via the operating software (Prospa) to compensate for the parasitic capacitance effect of the MOSFETs.

The operating frequency limit of the Q-switch design is mainly due to two physical properties of the MOSFET device, the total parasitic capacitance and the switching time. The total parasitic capacitance is the sum of the input, output, and reverse transfer capacitance of the device, 0.3 nF in this case. The switching time is the sum of the rise, fall, and reverse recovery time of the device, 52 ns in this case, which is equivalent to 19.2 MHz. While it is capable of operating at tens of Megahertz, consideration should be made on the device capacitance of 0.3 nF as it might be too big for NMR probes operating at MHz frequencies as that would offset the tuning frequency. NMR probes operating at above 10 MHz often have tuning capacitance values of under 1 nF depending on Q. The 0.3 nF from the device thus becomes significant in comparison. In future we will explore the use of smaller FETs with smaller gate-channel capacitance to increase performance and range.

The Q-switch was designed to be controlled directly from Prospa using a TTL digital logic line. This gives internal control and optimal timing on switching as Prospa also provides the pulse sequence to run the Terranova. The Terranova system has a convenient external port that provides logic outputs and which is readily configurable using Prospa. The Pulse & Collect (FID) pulse sequence was used in our experiments and was modified to have the logic output function enabled. Using logic lines to switch MOSFETs could introduce switching noise into the system as high frequency oscillations are common at

the on/off switching edges of a logic signal, therefore an isolator was used in the design. The isolator provides a noise barrier by using different ground connections for the input and output signals. While the isolator is good for this purpose, a drawback is the output logic signal does not provide enough current to switch the MOSEFTs rapidly on and off. A gate driver thus had to be used for both MOSFETs. An iCoupler series digital isolator from *Analog Devices* [14] was used in the design and a dual-output MOSFET driver from *intersil* [15] was used to drive the MOSFETs. A simplified block diagram of the design and the more detailed schematic (Altium Designer [16]) is shown in Figure 2.

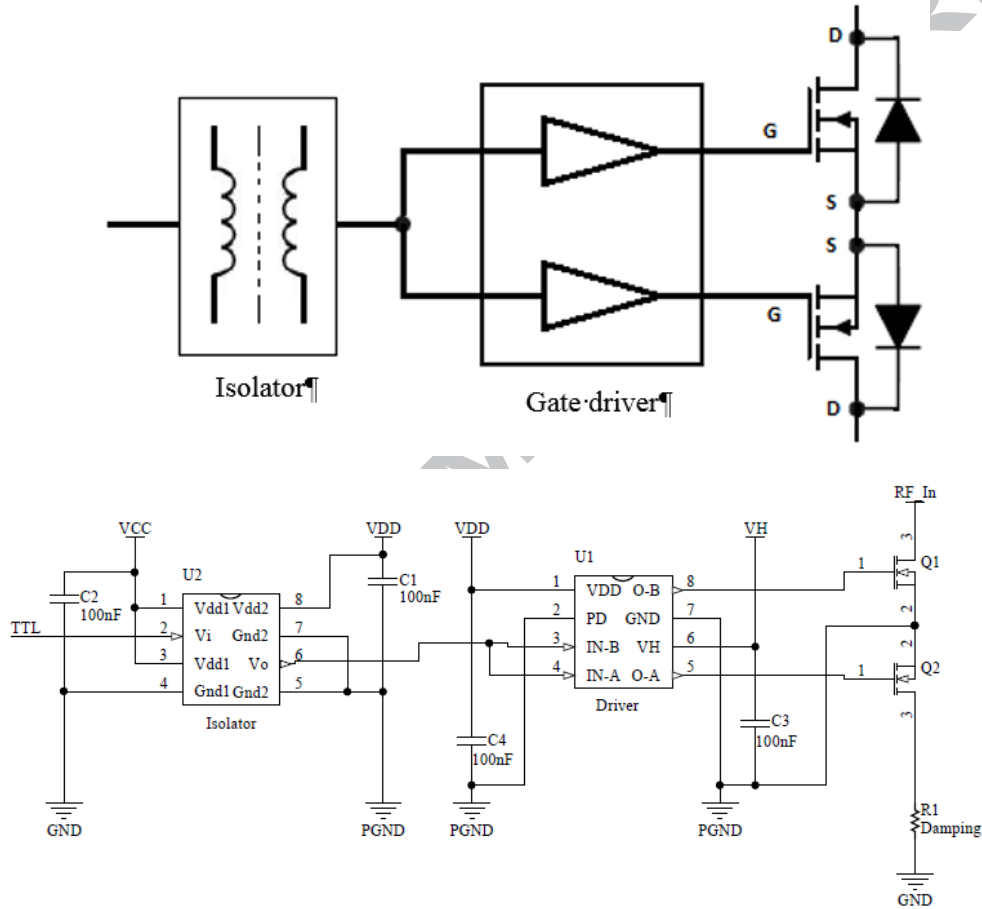


Figure 2. Block diagram (top) and schematic (bottom) showing the isolator, gate driver and MOSFET configuration for the Q-switch. Note the different reference point for the gate driver.

The ring-down from the original probe was first investigated using a 2 ms r.f. pulse. The response waveforms at different parts of the system are shown in Figure 3. On application of the r.f. pulse (a), the voltage across the B_1 coil (b) responded with the same waveform shape. However, at the end of the r.f. pulse there is ring-down from the coil. This ring-down effect is more evident at the output of the receiver (c). The receiver contains a high-gain amplifier, the r.f. pulse and the ring-down causes the receiver to saturate, resulting in the observed total recovery time of approximately 20 ms and thus the recommended minimum acquisition-delay time of 25 ms.

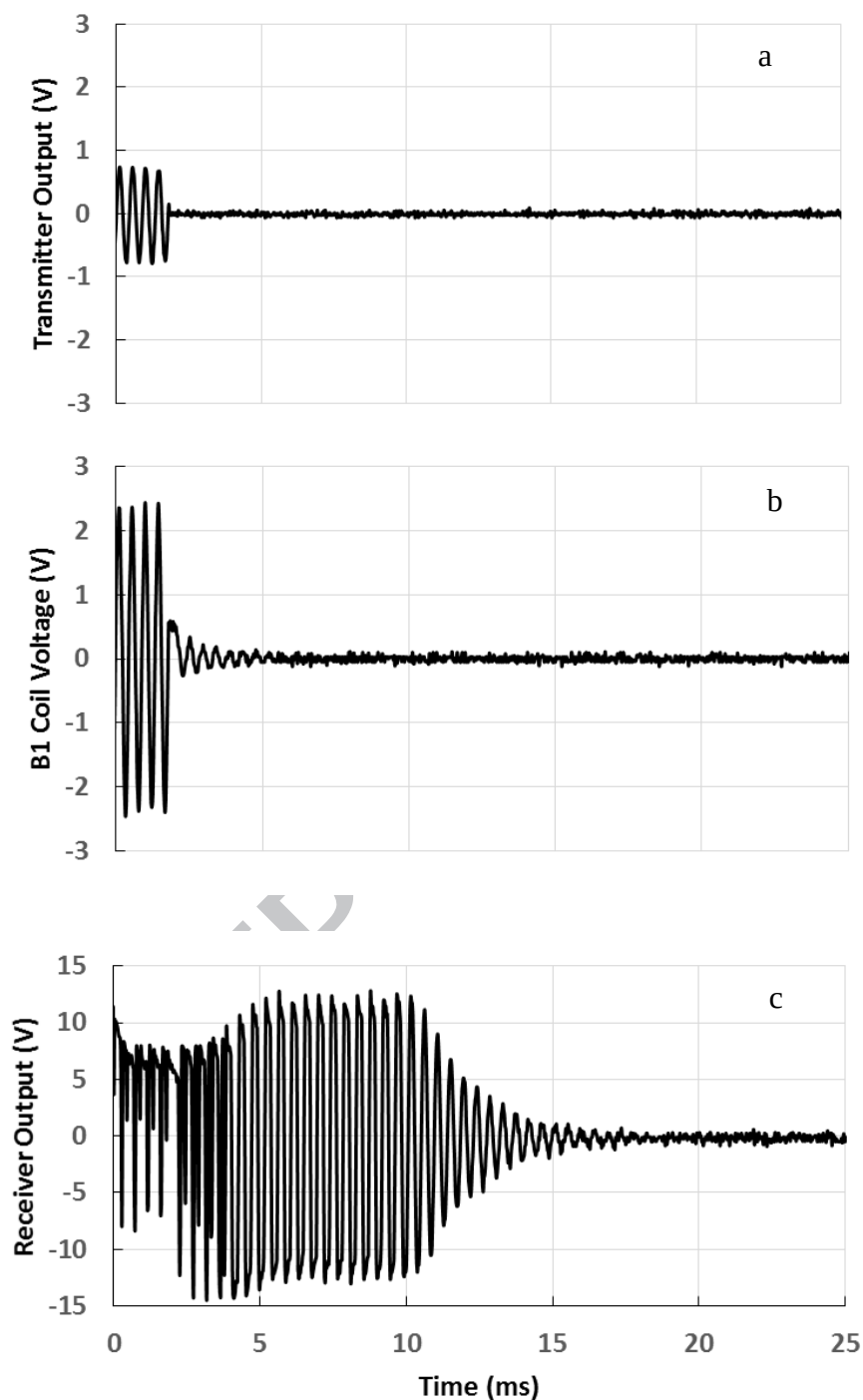


Figure 3. r.f. pulse and ring down effect of the B_1 coil (no test sample inside the coil). (a) is the r.f. pulse from the transmitter, (b) is the signal at the B_1 coil. There is clear evidence of the ring-down of the coil at the end of the r.f. pulse. (c) is the output of the receiver. It is clear that the receiver is saturated by the r.f. pulse resulting in a recovery time of ~ 20 ms.

The B_1 coil has an inductance value of 0.36 H, a resistance value of 322 ohms and a capacitance of 4.3 nF. A tuning capacitance of 9.6 nF was used in Prosipa, giving a combined capacitance value of 13.9 nF and a resonance frequency of 2.25 kHz. The Q of the coil was calculated to be 15.8. The damping resistance, R_d , was adjusted manually to obtain different values of Q in search of the best ring-down time, by observing the receiver output signal. Here we define the damping time as the time duration when the Q-switch is “on”, starting at the end of the r.f. pulse. The ring-down time is then assigned to be the time from the end of the r.f. pulse until the ring-down signal drops to the ambient noise level of approximately 4 μ V. A range of R_d values were tested with varying damping times. Table 1 shows sample data (for a damping time of 1 ms) of the damping resistance used, the corresponding Q values, the observed system dynamic and the resultant ring-down time respectively. Note that the Q of the system is changed only during the damping time. As soon as the Q-switch is off, the system Q returns to the original value of 15.8, for the acquisition of the required NMR signal. It was empirically found that the combination of $R_d = 2.15$ k Ω and a damping time of 1 ms gives the shortest ring down time of 8 ms. The ring-down waveforms are reproducible and the ring-down times are very consistent in each case, with an uncertainty of under 0.5 ms.

Table 1. Sample Ring Down time (τ) measurements

Damping resistor R_d (k Ω)	Q	System dynamic	Ring down time (ms)
0	15.8	Un-damped	20
1	3.8	Under-damped	16
2.15	2.1	Critically-damped	8
4.3	1.1	Over-damped	10

Figure 4 shows the corresponding ring-down waveforms for three damping scenarios corresponding to three different Q values. In Figure 4(b) for the under-damped scenario with a Q of 3.8, a significant ring down was still present after the 1 ms damping time, and hence the observed saturated output from the receiver. A damping time of longer than 1 ms is needed for this scenario. For Figure 4(c) the critically-damped scenario with a Q of 2.1, the ring down decays to the noise level within the damping time of 1 ms. However there is some oscillation introduced by turning off of the Q-switch. The oscilloscope used does not have the resolution to capture this microvolt level signal, but the receiver amplifier output clearly shows some ringing. For Figure 4(d), the over-damped scenario with a Q of 1.1, the oscillation from the switch-off decays away in about 0.5 ms, however there is over-shoot associated with over-damped systems which forces the receiver into saturation.

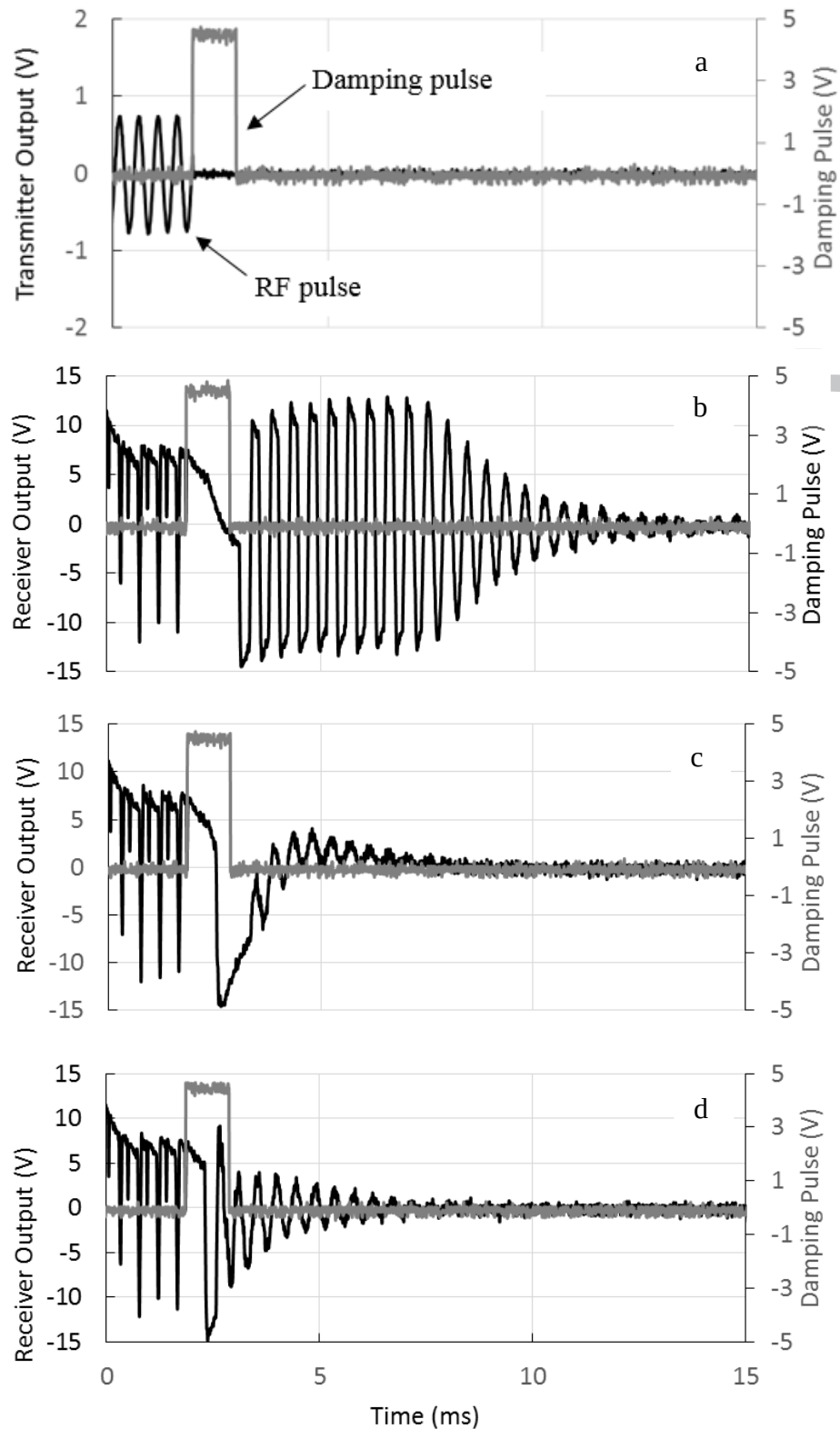


Figure 4. The effect of changing Q on the ring down time for under-damped, critically-damped, and over-damped systems. (a) shows the r.f. and the damping pulses; (b) is the ring down signal for $Q=3.8$ (under damped). (c) is the ring down signal for $Q=2.1$ (critically damped). (d) is the ring down signal for $Q=1.1$ (over damped). The best ‘recovery time’ is 8 ms for the critically damped system.

To compare the ring down times observed with theory, an estimate can be obtained using the exponential decay method [4]. By measuring the peak oscillation voltage of the B₁ coil at switch off (V_0), and using the ambient noise level as the steady state voltage ($V = 4 \mu\text{V}$), the decay time, t , can be calculated using equation 2, where τ is given by Equation 1, at 2.25 kHz and $Q = 15.8$ (Q-switch is off).

$$V/V_0 = e^{-t/\tau} \quad (2)$$

The results are shown in Table 2. The inclusion of the Q-switch did not introduce additional noise into the system. While the theoretical ring-down time values (Table 2) are not exactly the same as the measured values (Table 1), the trend shows good agreement.

Table 2. Calculated ring down time (τ) values.

System dynamic	Peak Voltage, V_0 , (mV)	Decay time, t (ms)	Damping Pulse (ms)	Ring-down time (ms) (Decay + Damping)
Un-damped	600	26.6	0	26.6
Under-damped	1	12.3	1	13.3
Critically-damped	0.05	5.6	1	6.6
Over-damped	0.2	8.7	1	9.7

The oscillation following turning off of the Q-switch was further investigated to see if the oscillation was from the Q-switch itself or as part of the ring down from the coil. This was done by switching on the critically damped Q-switch without the application of the r.f. pulse - ‘signal’ oscillation was observed as is shown in Figure 5. The oscillation is caused by the charge injection from the parasitic capacitance of the MOSFETs used in the Q-switch. However this oscillation decays away relatively fast and does not saturate the receiver as there is no energy stored in the B₁ coil. The amplitude of the oscillation is essentially the same as observed in the critically damped system above (Figure 4(c)), indicating that if this switch-induced oscillation can be eliminated the minimum ring down time and hence the acquisition delay could be reduced further. This is a limitation of the MOSFET technology as each device has a finite value of parasitic capacitance and gives a charge injection into the circuit when operating. This effect can be minimized by using new devices available in the market featuring several developments with respect to MOSFET technology [17-20].

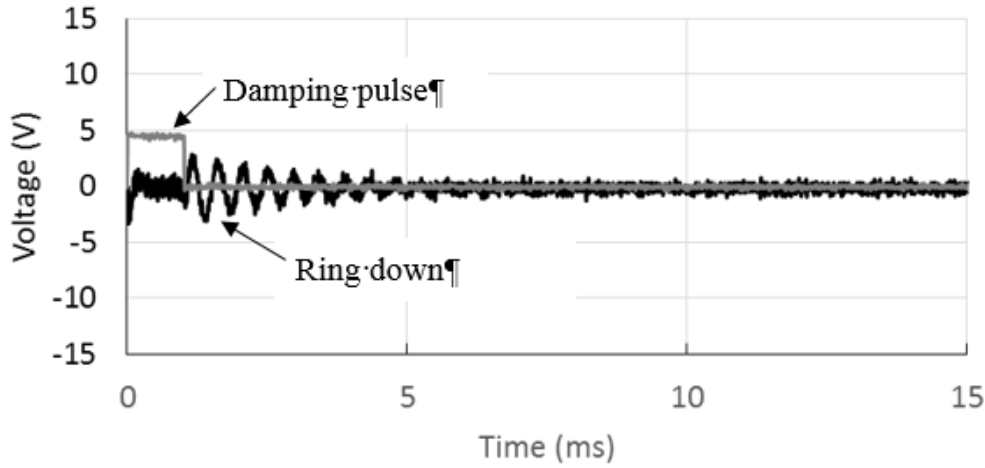


Figure 5. Oscillation introduced by the turn off of the Q-switch after the 1 ms damping time, without applying a r.f. pulse. Oscillation is observed after the Q-switch turn off. The vertical axis is consistent with Figure 4 for ease of comparison.

3. Results and Discussion

3.1. Free induction decay

Reducing the acquisition delay is demonstrated by measurements on a material featuring a fast effective spin-spin relaxation time (T_2^*), where the NMR signal can significantly diminish during the acquisition delay (without the Q-switch). The FID signal from canola oil (570 mL in a bottle) was measured with a simple pulse and collect (FID) experiment. The experimental acquisition delay (t_{delay}) is 9 ms when using the Q-switch and 25 ms without the Q-switch. Figure 6 shows a comparison of the FID signal of canola oil both with and without the Q-switch operating.

It is evident in Figure 6 that the FID signals are nearly identical 25 ms after the excitation pulse; the maximum signal detected increases from 24.5 μ V without the Q-switch to 30.8 μ V with the Q-switch (corresponding to a 26% increase in initial signal). The FID signals are processed and fit according to the following equation in order to estimate a single effective spin-spin relaxation constant (T_2^*) for each acquisition [21];

$$S(t) = S_0 \exp\left(-\frac{t_a + t_{delay}}{T_2^*}\right), \quad (3)$$

where S_0 is the initial magnetisation, t_a is the acquisition time and t_{delay} is the pulse to acquisition delay (t_a and t_{delay} are highlighted in Figure 6 for the two acquisitions with and without the Q-switch). The measured T_2^* times are 113 and 112 ms for the FID signals acquired with and without the Q-switch, respectively. The similarity in T_2^* constants verifies that the signals behave nearly identical (as

expected), however the purpose of this experiment is primarily to demonstrate the additional earlier signal which was possible to acquire with the Q-switch.

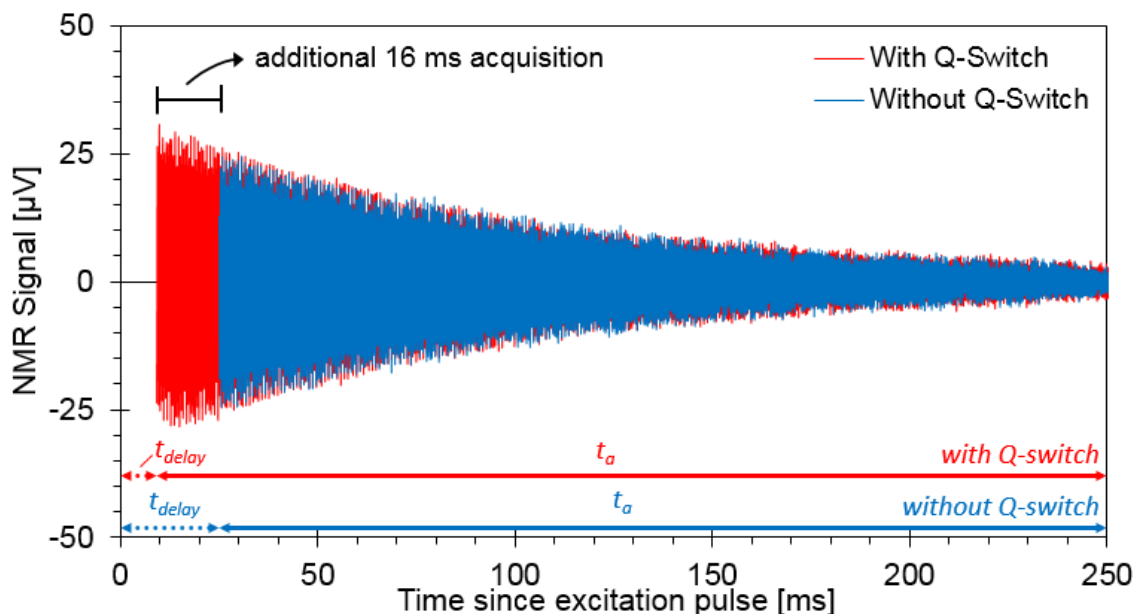


Figure 6. FID signal of a canola oil sample (in a 570 mL bottle). The FID signals are acquired with ($t_{\text{delay}} = 9$ ms) and without ($t_{\text{delay}} = 25$ ms) the Q-switch applied (red and blue lines respectively – blue is superimposed on red). The acquisition parameter times; t_{delay} and t_a , as well as the additional 16 ms of acquisition time which is possible to acquire due to the Q-switch are highlighted.

3.2. Signals with varying relaxation times

A potentially useful application of the Q-switch is now considered when examining materials at a range of relaxation times. This is demonstrated by analysing the relaxivity of a NMR doping agent at varying concentrations in water. The effect of iron oxide particle (Fe_3O_4 , 20–100 μm) concentration on the FID signal is examined both with and without the Q-switch. The concentration of iron oxide is logarithmically varied from 0.01 mMol/L to 10 mMol/L (9 different concentrations). The FID signals are fit with the model signal decay (Equation 3) using a Tikhonov regularisation inversion procedure^[22] to enable fitting of an appropriate T_2^* distribution ($P(T_2^*)$) to the experimental FID. The compromise between a consistent solution and a smooth solution is achieved by selecting a suitable regularisation parameter using the generalised cross-validation method^[23].

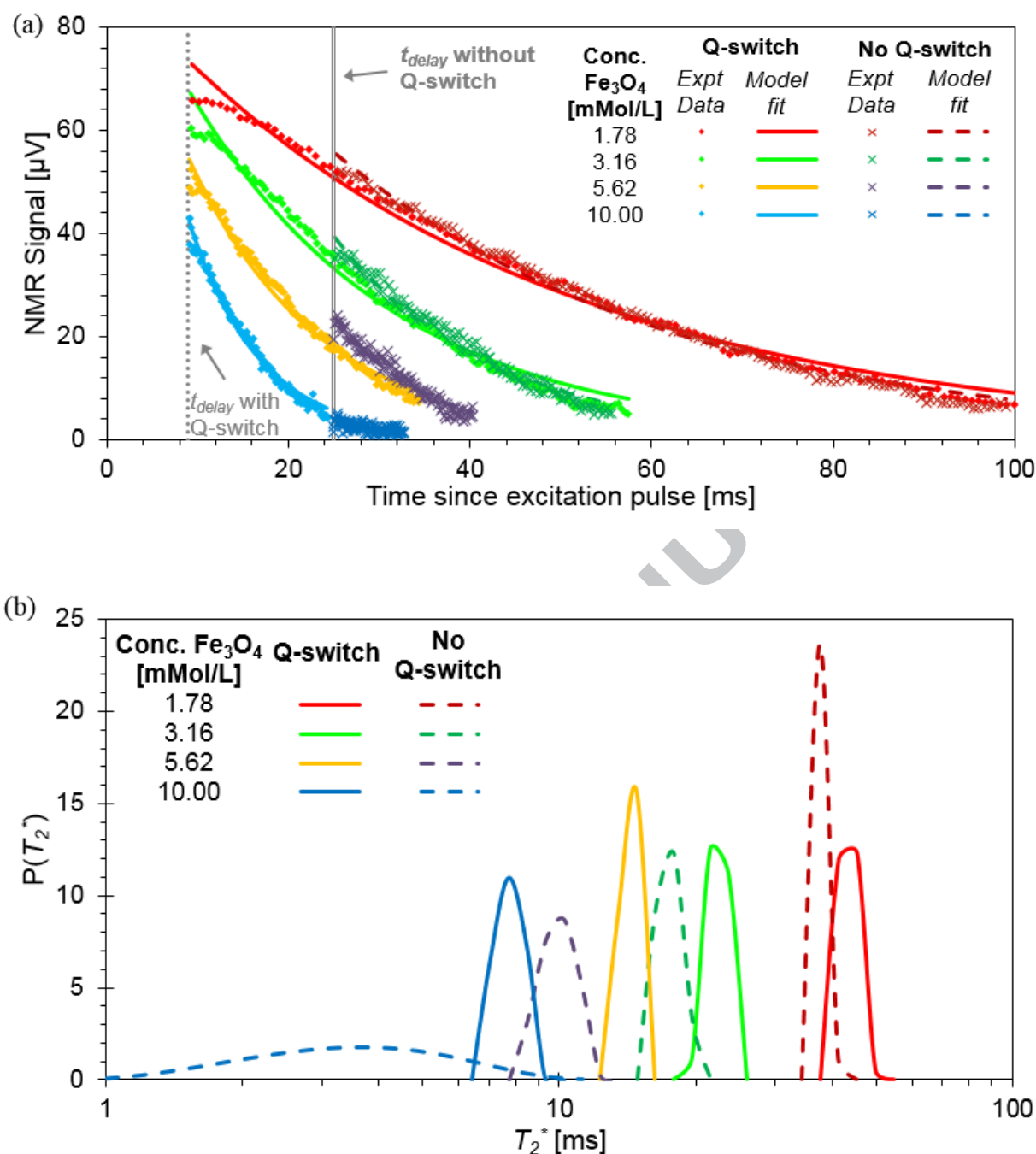


Figure 7. The T_2^* distributions determined by regularisation for the FID signals acquired both with and without the Q-switch at varying iron oxide concentrations in water. (a) The model fit of Equation 3 to the experimentally obtained signals using regularisation is displayed for each FID signal. The acquisition time delay (t_{delay}) with and without the Q-switch is indicated. (b) The resultant T_2^* probability distributions corresponding to the model fits displayed in (a).

Figure 7(a) shows the processed FID signal envelope as well as the regularised signal fit (of Equation 3) to the experimental FID data. The data is shown at comparatively high concentrations of iron oxide where the advantage of the Q-switch is more prevalent. The signal envelopes and model signal fits are shown for FID acquisition both with and without the Q-switch applied. Figure 7(b) displays the resultant T_2^* probability distributions ($P(T_2^*)$). The initial signal for un-doped water acquired with the Q-switch (70.0 μV) is used as a reference for the anticipated maximum signal (the signal immediately following the excitation pulse). From Figure 7 it can be observed that a significant proportion of the NMR signal is essentially lost during the first 25 ms of acquisition when a Q-switch is not used. For example, at an iron oxide concentration of 5.62 mMol/L, the first acquired point of the FID acquired with the Q-switch (48.9 μV) is $\sim 70\%$ of the maximum signal. The first point of the FID acquired without the Q-switch (19.4 μV) is only $\sim 28\%$ of the maximum signal.

In Figure 7(a) there is a potentially unexpected FID signal decay behavior during the first few points of acquisition when the Q-switch is applied. This can be attributed to non-exponential decay of the FID signal [24, 25]. The decay shape of the FID signal is dependent of the local magnetic field inhomogeneity distribution. When the field distribution is Lorentzian, then the corresponding FID signal will have an exponential decay. However, when the field distribution is Gaussian, then the corresponding FID signal will have a half-Gaussian shaped decay. This half-Gaussian shaped FID decay matches the observed signal behavior for the experimentally acquired FID signals in this work and will be implemented in future.

The inverted distributions ($P(T_2^*)$ - Figure 7(b)) acquired in the absence a Q-switch generally produce much broader distributions, particularly at higher iron oxide concentrations, reflecting a greater level of uncertainty in the data. This also manifests as erroneously low T_2^* values. The relative change in effective spin-spin relaxation rate $1/\Delta T_2^*|_c$ at an iron oxide concentration c is given by:

$$\frac{1}{\Delta T_2^*|_c} = \frac{1}{T_2^*|_c} - \frac{1}{T_2^*|_0} \quad (4)$$

where $T_2^*|_c$ is the effective spin-spin time constant at an iron oxide concentration c and $T_2^*|_0$ is the effective spin-spin time constant for un-doped water. Figure 8 shows the change in mean $1/\Delta T_2^*|_c$ as a function of iron oxide concentration.

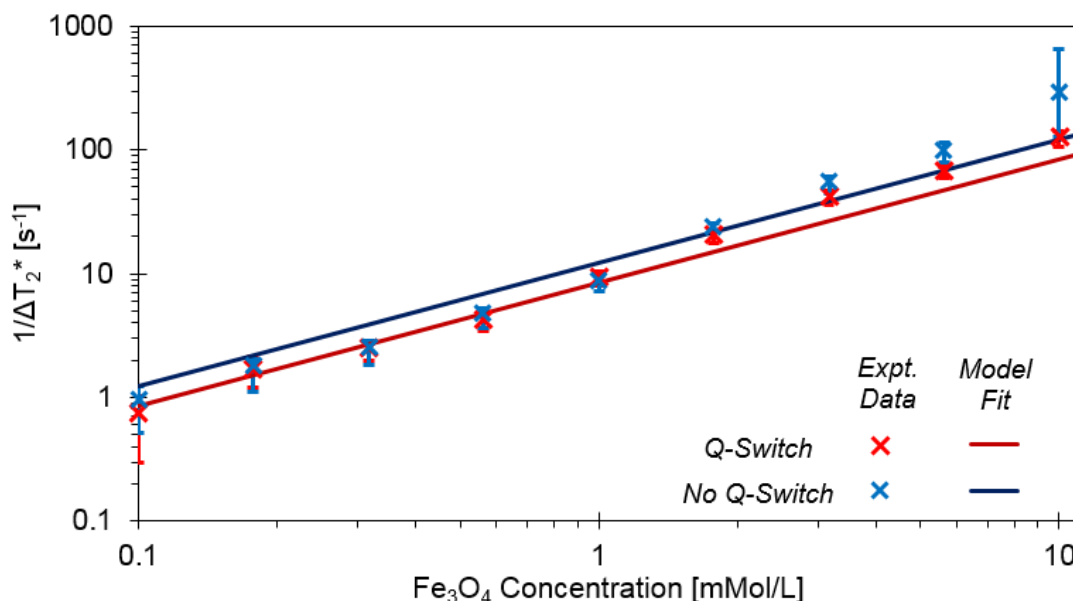


Figure 8. Change in the mean spin-spin relaxation rate ($1/\Delta T_2^*$) as a function of the concentration of iron oxide particles in water. The results from FID signals acquired using the Q-switch are indicated by red diamonds ($\square$) and the values determined from FID signals acquired without the Q-switch are indicated with blue crosses ($\times$). Error bars indicate 95% confidence intervals determined from the regularised probability distribution. The linear fit of the two sets of data as per Equation 5 is represented by the solid lines.

From Figure 8, it is observable that there is a good correlation between the two measured data sets (with and without the Q-switch) at low concentrations (≤ 1 mMol/L) due to minimal signal loss for both data sets. However at higher concentrations (> 1 mMol/L) the difference in measurements becomes more pronounced. The relaxivity of the iron oxide particles (r_2^*) can be determined by applying a linear regression to the measured effective relaxation rate ($1/\Delta T_2^*$) versus concentration data [26]. The relationship is represented by the following equation;

$$\frac{1}{\Delta T_2^*} = r_2^*[C(Fe_3O_4)] \quad (5)$$

The linear fits are shown on Figure 8. The relaxivity is $10.5 \text{ L mMol}^{-1} \text{ s}^{-1}$ (with a standard error of fit of 65 %) when fitted to the data acquired using the Q-switch whilst the relaxivity is estimated as $12.2 \text{ L mMol}^{-1} \text{ s}^{-1}$ (with a standard error of fit of 164 %) for data acquired without the Q-switch. The deviation between estimates is attributed to the significant initial signal loss at higher iron oxide concentrations for the data acquired without a Q-switch. The above data demonstrates that we can have more confidence in the analysis of faster decaying FID signals due to the earlier acquisition of the FID signal.

3.3. Q-switch applied to flow metering apparatus

We have previously demonstrated an application of the EFNMR system as a flow metering apparatus [^{8,9}]. The flow measurement methodology (as outlined in detail in O'Neill *et al.* [⁹]) essentially uses the FID signal in order to determine the velocity distribution of the flowing stream. It is critical to obtain an as accurate as possible acquisition of the full FID signal in order to determine the relevant velocity distribution. The measured NMR signal is a composite of three contributions: signal accumulation during polarisation (S_P), signal attenuation from intermediate decay between the polarisation magnet and the EFNMR r.f. detector (S_{PD}), and signal attenuation following excitation (S_D). The measured FID signal (S) of a fluid element flowing (at velocity v) through the EFNMR detection coil is modelled by

$$S(v, t_a) = S_0 \left[1 - e^{-\frac{L_P}{vT_1}} \right] \left[e^{-\frac{L_{PD}}{vT_1}} \right] \left[1 - \frac{(t_{delay} + t_a)v}{L_D} \right] e^{-\frac{t_{delay} + t_a}{T_2^*}} \quad \text{for } t_{delay} + t_a \leq \frac{L_D}{v}, \quad (6)$$

where t_a is the time period during signal acquisition, S_0 is the NMR signal after an infinite time in the magnetic field, L_P is the polarisation magnet length, L_{PD} is the polarisation-detection separation distance, L_D is the detection coil length, T_1 is the spin-lattice relaxation time, T_2^* is the effective spin-spin relaxation time and t_{delay} is the acquisition delay between excitation and detection. The above model (Equation 6) is fit to experimentally obtained FID signals using Tikhonov regularisation using the procedure discussed in O'Neill *et al.* [⁹] in order to determine the probability distribution of v .

A range of NMR flowmeter experiments were conducted in order to demonstrate the improvement in accuracy for the flow measurement apparatus with the inclusion of the Q-switch. Water flows through the flow loop at flowrates of 1–9 m³/h (corresponding to mean water velocities ($V_{M,rotameter}$) of 0.37–3.31 m/s). Figure 9(a) shows the experimentally acquired FID signals of the flowing stream, both with and without the Q-switch (such that $t_{delay} = 9$ ms and 25 ms respectively) for mean velocities of 0.37 m/s to 3.31 m/s. Figure 9(b) shows the corresponding resultant velocity probability distributions, $P(v)$.

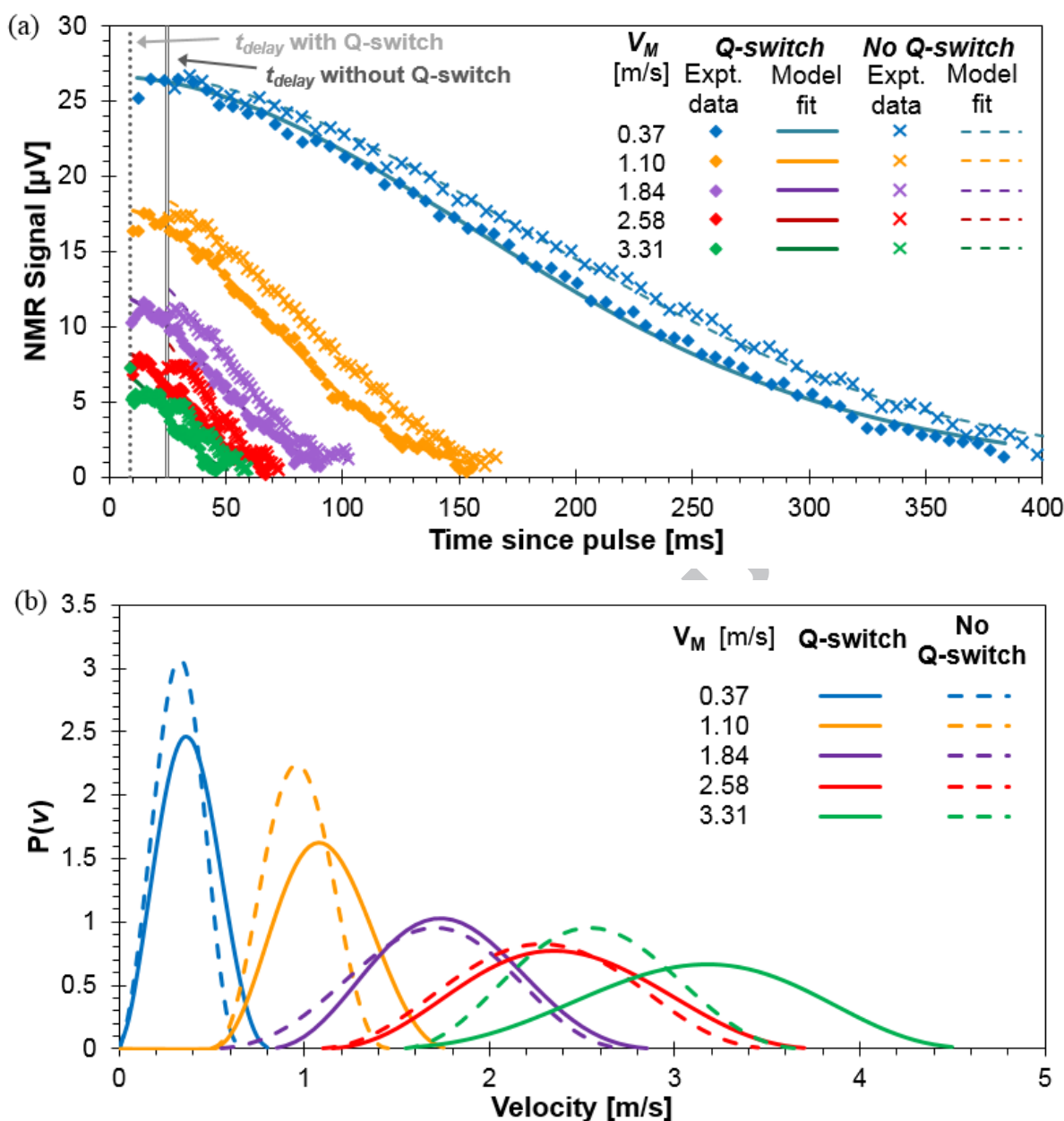


Figure 9. (a) Equation 6 fit to experimentally acquired FID signals of a flowing stream via Tikhonov regularisation. Water is flowing at mean velocities of 0.37 – 3.31 m/s. Experimental FID signals are acquired and fit both with and without the Q-switch. The acquisition delay time (t_{delay}) with and without the Q-switch is indicated. (b) The corresponding velocity probability distributions at each mean flow velocity.

The regularised model fits are observed to fit reasonably well for all experimental signals, as expected. At the low velocity (0.37 m/s) the measured velocity probability distributions (both with and without the Q-switch) are relatively similar. As the measured velocity increases, a significant difference between the measured distributions can be observed in Figure 9. This is explored further by examining

variations in the mean velocity of each distribution with and without a Q-switch. Figure 10(a) shows the mean NMR-measured velocity ($V_{M,NMR}$) compared to the mean velocity as measured by the in-line rotameter ($V_{M,Rotameter}$), Figure 10(b) shows the corresponding measurement deviation.

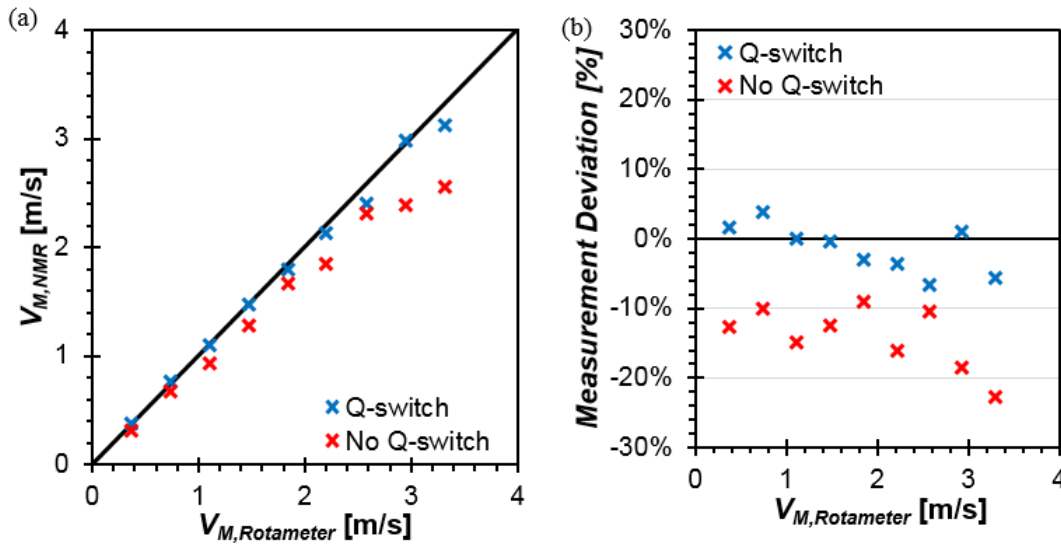


Figure 10. Comparison of NMR predicted mean flow velocity ($V_{M,NMR}$), both with and without the Q-switch applied, to the independent rotameter velocity measurement ($V_{M,Rotameter}$). (b) The relative measurement deviations between $V_{M,NMR}$ and $V_{M,Rotameter}$.

The NMR predicted mean velocity measurements (both with and without the Q-switch) agree relatively well with the in-line rotameter measurement of the mean velocity at lower velocities ($V_M < 1.0$ m/s). However at higher velocities ($V_M > 1.0$ m/s) there is a greater discrepancy for the data acquired without a Q-switch. The NMR signal from high velocity components decay faster (relative to slower moving components) due to the ‘flush-out’ effect of excited fluid leaving the NMR detection region. This results in an effective signal loss during the initial FID acquisition time, which is not measured in the absence of a Q-switch. The Q-switch serves to partially alleviate this limitation by reducing the acquisition delay time.

To further demonstrate the potential improvement the Q-switch provides for the flow metering apparatus, Figure 11 shows the effective signal loss during the acquisition delay, modelled using Equation 6, as a function of the mean fluid velocity. Experimental measurements of this signal loss are also shown in Figure 11:

$$\text{Signal loss } (t_{\text{delay}}) = \frac{S(v,0) - S(v,t_{\text{delay}})}{S(v,0)} \quad (7)$$

where $S(v, t_{\text{delay}})$ is the first point of the experimental FID measurement at a mean velocity v and $S(v, 0)$ is the signal at time zero determined from the regularized fit of Equation 6 to the same FID measurement. The experimentally measured signal loss is observed to match the model estimated signal loss reasonably well. Deviations at higher velocities for the data acquired without the Q-switch can be attributed to the incorrect extrapolation to $S(v, 0)$ as a result of an inaccurate velocity measurement. Assuming a signal loss of $< 10\%$ is acceptable, the Q-switch has increased the applicable range of the flow meter from ~ 1 m/s to ~ 3.3 m/s. This is a very considerable increase in the systems measurement range and importantly now allows it to measure industrially applicable fluid flow rates (typically 1-3 m/s).

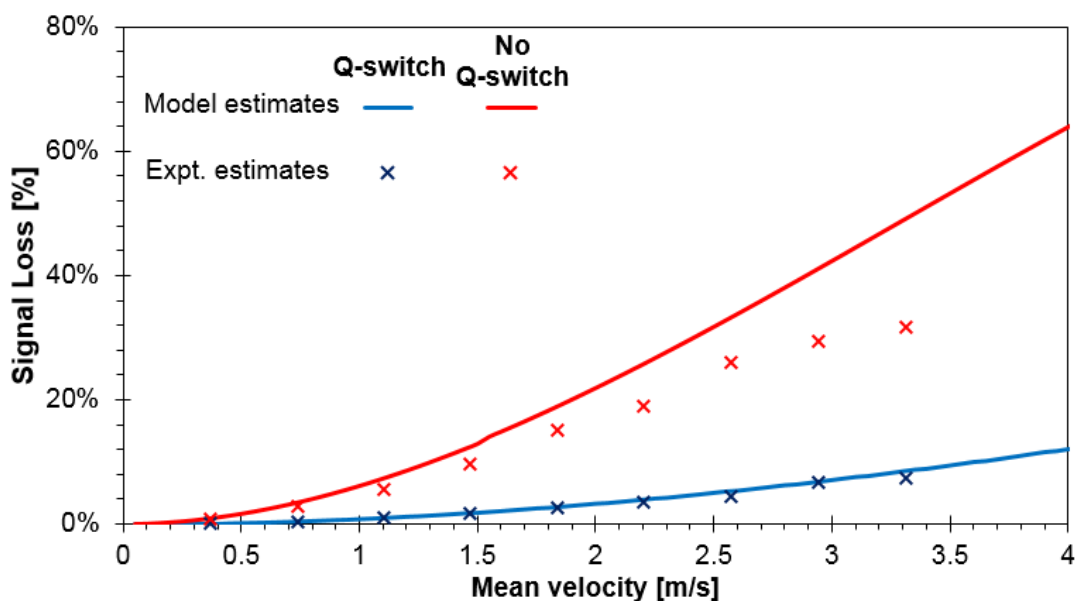


Figure 11. Comparison of the effective signal loss during the initial acquisition delay (t_{delay}) as a function of fluid velocity both with the Q-switch ($t_{\text{delay}} = 9$ ms) and without the Q-switch ($t_{\text{delay}} = 25$ ms). Experimental measurements of signal loss are shown relative to the modelled signal loss calculated from Equation 6.

4. Conclusion

A NMR Q-switch was successfully designed and implemented external to a commercial low-field NMR spectrometer (which detected ^1H NMR signal at the Earth's magnetic field), although it is readily transferable to other spectrometers. This exploited a relatively simple resistive damping design featuring a double MOSFET switch. The design was optimized by adjusting the damping time and resistance value and resulted in a reduction in the signal acquisition delay from 25 ms to 9 ms. The

earlier acquisition was able to more reliably describe the relationship between measured T_2^* and the concentration of iron oxide particles in solution. It was also demonstrated using a NMR Flow meter, which we have previously reported on [8-10] and showed much better agreement with an independent rotameter measurement over a wider range of velocities.

5. Acknowledgments

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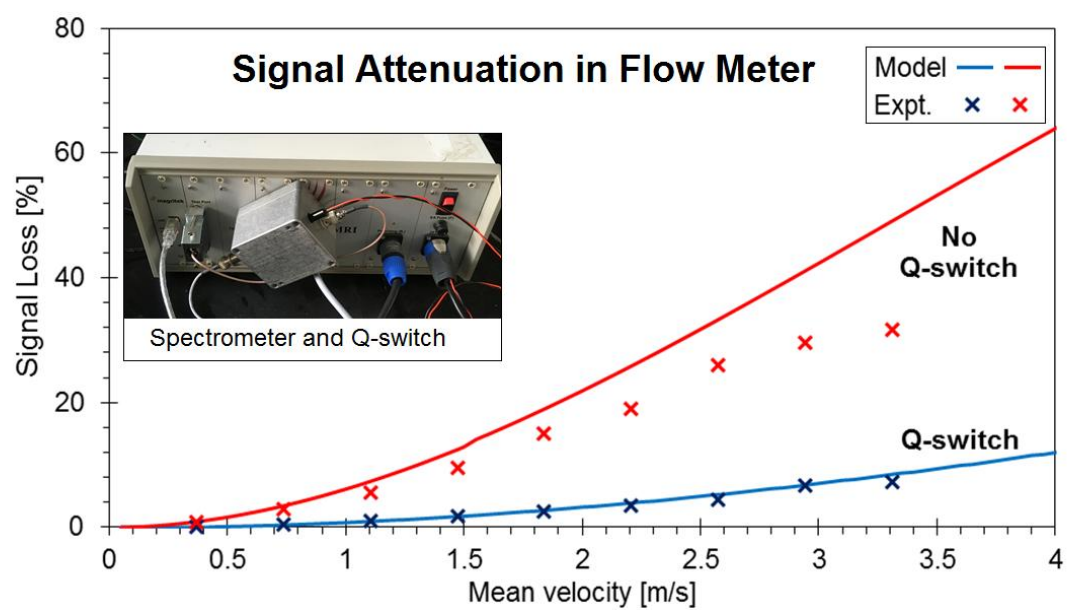
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- A resistive Q-switch for low field NMR systems operating up to 2 MHz is proposed
- No passive tuning elements and operating frequency only limited by MOSFET devices
- Q-switch allows earlier signal acquisition and extends NMR flow-meter range
- Gained 16 ms of data acquisition on Earth's field NMR system

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¹ These authors have contributed equally.